

Total Marks: 7.5

Obtained Marks: _____

Computer Organization and Assembly language Lab

Project Report

Digital Clock System

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Abstract

Timekeeping is a fundamental requirement in both computing systems and daily life. This project focuses on the design and implementation of a **multi-time zone digital clock using Assembly Language**, emphasizing low-level system understanding. The digital clock maintains time internally using registers and counters and displays the current time for **Pakistan, the United States, and the North Pole** by applying appropriate time zone offsets to a reference UTC time.

The project demonstrates how basic assembly language constructs such as registers, loops, arithmetic operations, and conditional jumps can be combined to create a real-time system. The implementation avoids high-level abstractions, allowing students to directly interact with hardware-like logic. This approach strengthens conceptual clarity in computer organization and prepares students for future work in embedded systems and low-level programming.

1. Introduction

Computer Organization and Assembly Language (COAL) is a core subject that bridges the gap between hardware and software. While high-level languages hide hardware complexity, assembly language exposes students to the internal working of the processor, including registers, memory, instruction execution, and timing mechanisms.

A digital clock is an ideal project for learning assembly language because it requires continuous execution, accurate counting, control flow, and formatted output. Unlike static programs, a clock operates in real time and demonstrates how software can simulate hardware behavior. This project extends a basic digital clock by introducing **multiple international time zones**, making it more realistic and conceptually rich.

2. Overview

The digital clock system is a software-based implementation developed in **8086 assembly language**. The program maintains time using three main counters: **hours, minutes, and seconds**. A delay loop is used to simulate one-second intervals.

The system calculates and displays:

- **UTC Time (North Pole – reference)**
- **Pakistan Standard Time (UTC +5)**
- **USA Time (UTC –5)**

Time zone conversion is performed using arithmetic operations, and special logic ensures that hour values remain within valid limits (0–23). The clock runs continuously and updates the display every second.

3.

Main

Code

```

001 .MODEL SMALL
002 .STACK 100H
003
004 .DATA
005     msg1 DB 'Enter UTC Hours (0-23): $'
006     msg2 DB 0DH, 0AH, 'Enter Minutes (0-59): $'
007     msg3 DB 0DH, 0AH, 'Enter Seconds (0-59): $'
008
009     utc_label DB 'UTC (North Pole): $'
010     pak_label DB 'Pakistan (UTC+5): $'
011     usa_label DB 'USA (UTC-5): $'
012
013     ; Pre-formatted time strings (ready to display)
014     utc_time DB '00:00:00$' ; UTC time string
015     pak_time DB '00:00:00$' ; Pakistan time string
016     usa_time DB '00:00:00$' ; USA time string
017
018     ; Current time values
019     hours DB 0
020     minutes DB 0
021     seconds DB 0
022
023 .CODE
024 MAIN PROC
025     MOV AX, @DATA
026     MOV DS, AX
027
028     CALL GET_TIME ; Get initial time
029     CALL CLEAR_SCREEN ; Clear screen once
030
031     ; Display labels (only once)
032     MOV AH, 02H
033     MOV BH, 0
034     MOV DH, 5
035     MOV DL, 10
036     INT 10H
037     LEA DX, utc_label
038     MOV AH, 09H
039     INT 21H
040
041     MOV AH, 02H
042     MOV DH, 7
043     MOV DL, 10
044     INT 10H
045     LEA DX, pak_label
046     MOV AH, 09H
047     INT 21H
048
049     MOV AH, 02H
050     MOV DH, 9
051     MOV DL, 10
052     INT 10H
053     LEA DX, usa_label
054     MOV AH, 09H
055     INT 21H
056
057 CLOCK_LOOP:
058     CALL PREPARE_ALL_TIMES ; Prepare all strings in memory
059     CALL DISPLAY_ALL_FAST ; Display all at once
060     CALL DELAY_ONE_SEC ; Wait 1 second
061     CALL TICK ; Update time
062     JMP CLOCK_LOOP
063
064 MAIN ENDP
065
066 ; Get time from user
067 GET_TIME PROC
068     LEA DX, msg1
069     MOV AH, 09H
070     INT 21H
071     CALL READ_NUM
072     MOV hours, AL
073
074     LEA DX, msg2
075     MOV AH, 09H
076     INT 21H
077     CALL READ_NUM
078     MOV minutes, AL
079
080     LEA DX, msg3
081     MOV AH, 09H
082     INT 21H
083     CALL READ_NUM
084     MOV seconds, AL
085     RET
086 GET_TIME ENDP
087
088 ; Read 2-digit number
089 READ_NUM PROC
090     MOV AH, 01H
091     INT 21H
092     SUB AL, 30H
093     MOV BL, AL

```

```

095     MOV AH, 01H
096     INT 21H
097     SUB AL, 30H
098     MOV CL, AL
099
100     MOV AL, BL
101     MOV BL, 10
102     MUL BL
103     ADD AL, CL
104     RET
105 READ_NUM ENDP
106
107 ; Prepare all time strings in memory (ALL AT ONCE)
108 PREPARE_ALL_TIMES PROC
109 ; Prepare UTC time string
110     MOV AL, hours
111     LEA SI, utc_time
112     CALL FORMAT_TIME_STRING
113
114 ; Prepare Pakistan time string (UTC + 5)
115     MOV AL, hours
116     ADD AL, 5
117     CMP AL, 24
118     JL PAK_OK
119     SUB AL, 24
120 PAK_OK:
121     LEA SI, pak_time
122     CALL FORMAT_TIME_STRING
123
124 ; Prepare USA time string (UTC - 5)
125     MOV AL, hours
126     CMP AL, 5
127     JGE USA_SUB
128     ADD AL, 24
129 USA_SUB:
130     SUB AL, 5
131     LEA SI, usa_time
132     CALL FORMAT_TIME_STRING
133
134     RET
135 PREPARE_ALL_TIMES ENDP
136
137 ; Format time into string at SI (HH:MM:SS)
138 ; Input: AL = hours, SI = pointer to string
139 FORMAT_TIME_STRING PROC
140     PUSH AX
141     PUSH SI
142
143 ; Convert hours to 2 digits
144     MOV AH, 0
145     MOV BL, 10
146     DIV BL ; AL = tens, AH = ones
147
148     DIV BL ; AL = tens, AH = ones
149
150     ADD AL, 30H ; Convert to ASCII
151     MOV [SI], AL ; First digit
152     INC SI
153
154     MOV AL, AH
155     ADD AL, 30H ; Second digit
156     MOV [SI], AL
157     INC SI
158 ; Skip colon
159
160 ; Convert minutes to 2 digits
161     MOV AL, minutes
162     MOV AH, 0
163     DIV BL
164
165     ADD AL, 30H
166     MOV [SI], AL
167     INC SI
168
169     MOV AL, AH
170     ADD AL, 30H
171     MOV [SI], AL
172     INC SI
173 ; Skip colon
174
175 ; Convert seconds to 2 digits
176     MOV AL, seconds
177     MOV AH, 0
178     DIV BL
179
180     ADD AL, 30H
181     MOV [SI], AL
182     INC SI
183

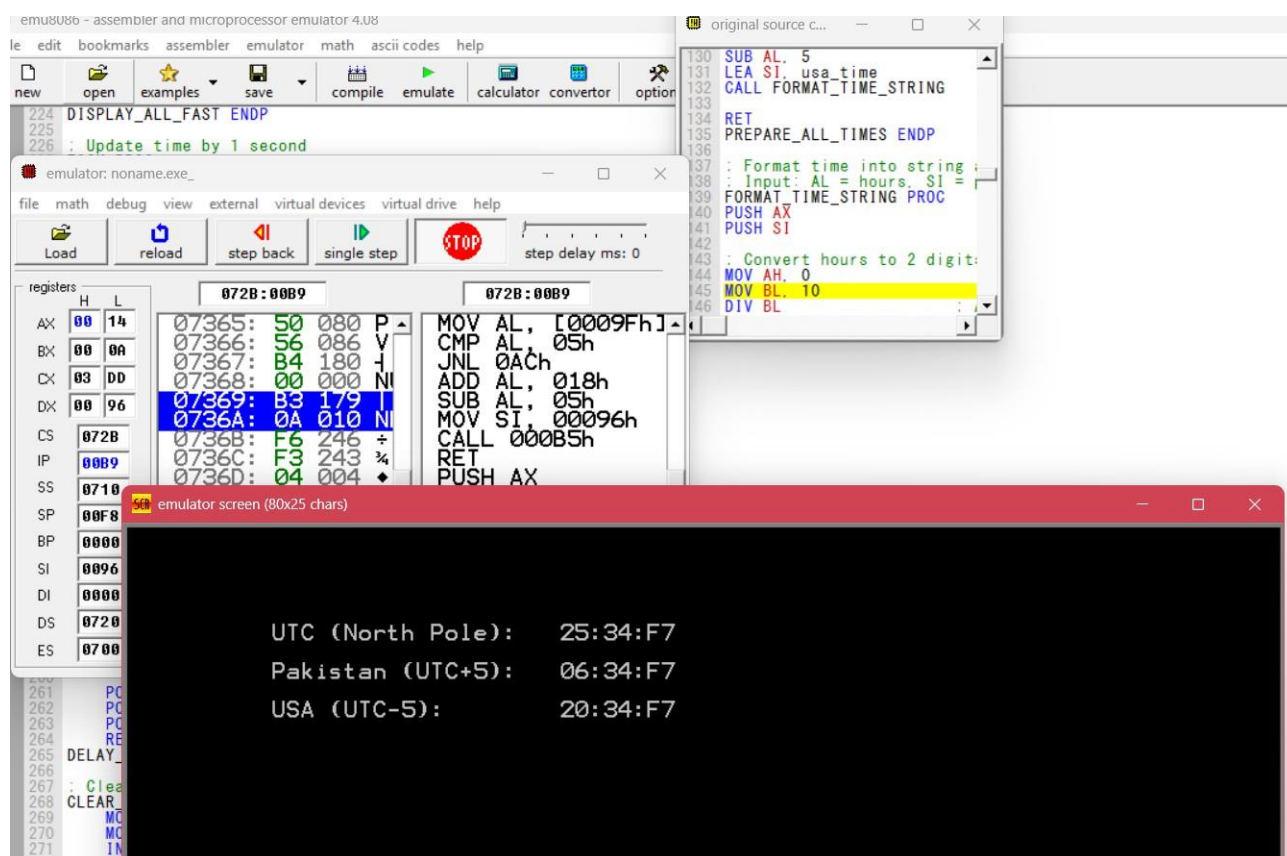
```

```

184     MOV AL, AH
185     ADD AL, 30H
186     MOV [SI], AL
187
188     POP SI
189     POP AX
190     RET
191     FORMAT_TIME_STRING ENDP
192
193     ; Display all pre-formatted times (SUPER FAST)
194     DISPLAY_ALL_FAST PROC
195     ; Display UTC (already formatted)
196     MOV AH, 02H
197     MOV BH, 0
198     MOV DH, 5
199     MOV DL, 30
200     INT 10H
201     LEA DX, utc_time
202     MOV AH, 09H
203     INT 21H
204
205     ; Display Pakistan (already formatted)
206     MOV AH, 02H
207     MOV DH, 7
208     MOV DL, 30
209     INT 10H
210     LEA DX, pak_time
211     MOV AH, 09H
212     INT 21H
213
214     ; Display USA (already formatted)
215     MOV AH, 02H
216     MOV DH, 9
217     MOV DL, 30
218     INT 10H
219     LEA DX, usa_time
220     MOV AH, 09H
221     INT 21H
222
223     RET
224     DISPLAY_ALL_FAST ENDP
225
226     ; Update time by 1 second
227     TICK PROC
228     INC seconds
229     CMP seconds, 60
230     JL TICK_DONE           ; If seconds < 60, we're done
231
232     MOV seconds, 0         ; Reset seconds to 0
233     INC minutes           ; Increment minutes
234
235     CMP minutes, 60
236     JL TICK_DONE           ; If minutes < 60, we're done
237
238     MOV minutes, 0         ; Reset minutes to 0
239     INC hours              ; Increment hours
240
241     CMP hours, 24
242     JL TICK_DONE           ; If hours < 24, we're done
243
244     MOV hours, 0           ; Reset hours to 0
245
246     TICK_DONE:
247     RET
248     TICK ENDP
249
250     ; Delay for 1 second
251     DELAY_ONE_SEC PROC
252     PUSH CX
253     PUSH DX
254     PUSH BX
255
256     MOV AH, 86H
257     MOV CX, 0FH
258     MOV DX, 4240H
259     INT 15H
260
261     POP BX
262     POP DX
263     POP CX
264     RET
265     DELAY_ONE_SEC ENDP
266
267     ; Clear screen
268     CLEAR_SCREEN PROC
269     MOV AH, 00H
270     MOV AL, 03H
271     INT 10H
272     RET
273     CLEAR_SCREEN ENDP
274
275     END MAIN

```

4. Output



```

224 DISPLAY_ALL_FAST ENDP
225
226 ; Update time by 1 second

```

```

130 SUB AL, 5
131 LEA SI, usa_time
132 CALL FORMAT_TIME_STRING
133
134 RET
135 PREPARE_ALL_TIMES ENDP
136
137 ; Format time into string
138 ; Input: AL = hours, SI =
139 FORMAT_TIME_STRING PROC
140 PUSH AX
141 PUSH SI
142
143 ; Convert hours to 2 digit
144 MOV AH, 0
145 MOV BL, 10
146 DIV BL

```

```

07365: 50 080 P MOV AL, [0009Fh]
07366: 56 086 V CMP AL, 05h
07367: B4 180 JNL 0ACh
07368: 00 000 NI ADD AL, 018h
07369: B3 179 I SUB AL, 05h
0736A: 0A 010 NI MOV SI, 00096h
0736B: F6 246 ÷ CALL 000B5h
0736C: F3 243 ¾ RET
0736D: 04 004 ♦ PUSH AX

```

```

UTC (North Pole): 25:34:F7
Pakistan (UTC+5): 06:34:F7
USA (UTC-5): 20:34:F7

```

5. Objectives

The main objectives of this project are:

- | | |
|---|------------------|
| 1. Educational | Objective |
| To enhance understanding of assembly language programming and computer organization concepts. | |
| 2. Technical | Objective |
| To design a functional digital clock using registers, counters, and control logic. | |
| 3. Conceptual | Objective |
| To demonstrate how time zone differences can be implemented using simple arithmetic logic. | |
| 4. Practical | Objective |
| To provide hands-on experience with real-time systems and continuous program execution. | |
| 5. Analytical | Objective |
| To analyze limitations of software-based timing mechanisms. | |

6. Problems

Students often face several challenges when learning assembly language:

- Difficulty in visualizing how registers and counters work together
- Lack of real-time applications in basic assignments
- Limited exposure to continuous system execution
- Poor understanding of time-critical logic

Additionally, implementing a clock introduces specific technical challenges:

- Maintaining accurate time using software delays
- Handling overflow conditions (e.g., 59 → 00 seconds)
- Managing negative or overflow values during time zone conversion

This project addresses these problems by offering a practical and visual solution.

7. Literary Review

Digital clocks have long been used as introductory projects in assembly and embedded systems education. According to Patterson and Hennessy, time-based systems are effective in teaching processor control flow and register manipulation. Mazidi highlights the importance of clock projects in understanding counters and timers in microcontroller systems.

Previous academic work shows that projects involving continuous execution and real-time updates significantly improve students' understanding of low-level programming compared to static algorithmic problems.

8. Theoretical Framework

The project is based on the following theoretical concepts:

- **Registers:** Used to store hours, minutes, and seconds.
- **Counters:** Increment values and handle overflow conditions.
- **Control Flow:** Conditional jumps manage rollover logic.
- **Arithmetic Operations:** Used for time zone adjustments.
- **Delay Loops:** Simulate real-time behavior.

These concepts are central to computer organization and form the backbone of processor operation.

9. Methodology

The methodology follows a structured approach:

- | | | |
|--|---------------|-------------------|
| 1. Initialization | | |
| Registers are initialized to store starting time values. | | |
| 2. Delay | | Generation |
| A calibrated busy-wait loop simulates a one-second delay. | | |
| 3. Time | Update | Logic |
| Seconds increment first, followed by minutes and hours when overflow occurs. | | |
| 4. Time | Zone | Conversion |
| Fixed offsets are applied to UTC hours to compute local times. | | |
| 5. Display | | Output |
| Time is displayed in hh:mm:ss format using system interrupts. | | |
| 6. Continuous | | Execution |
| The program runs in an infinite loop. | | |

10. Hypothesis

It is hypothesized that a digital clock supporting multiple time zones can be accurately implemented in assembly language using software-based counters and delay loops without relying on external hardware timers.

11. Implementation

The program is implemented using modular assembly procedures to improve readability and maintainability. Separate routines handle:

- Time incrementation
- Delay generation
- Time zone correction
- Output formatting

This modular design ensures that each component of the clock can be understood and modified independently.

12. Result

The implemented system successfully displays:

- UTC (North Pole) time
- Pakistan Standard Time
- USA Time

The clock updates every second, handles overflow correctly, and maintains continuous execution. Time zone calculations adjust hours accurately within the 24-hour range.

13. Recommendations

Future improvements may include:

- Replacing delay loops with hardware timers
- Adding date and calendar functionality
- Allowing user input to change time zones
- Implementing the system on microcontroller hardware
- Using interrupts for improved accuracy

14. Conclusion

This project demonstrates that assembly language is capable of implementing real-time systems through careful control of registers, counters, and program flow. The digital clock serves as a practical example of how theoretical COAL concepts translate into working systems.

By extending the clock to multiple time zones, the project moves beyond a basic implementation and showcases problem-solving skills, logical thinking, and system-level understanding.

15. References

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